

SSH Brute-Force Detection and Analysis Lab

Executive Summary

Problem / Objective

SSH brute-force/password-guessing detection. This project simulates an SSH password-guessing attack in a controlled lab environment and demonstrates how the resulting authentication activity can be collected, detected, and analyzed from a SOC perspective.

Lab Environment

A three-VM attack and detection lab designed to simulate malicious SSH authentication activity, collect the resulting logs, and analyze the activity using Splunk.

Kali Linux - Attacker: Hosts Metasploit and generates the simulated attack activity.

Metasploitable 2 - Target: Provides the vulnerable SSH service and generates authentication logs.

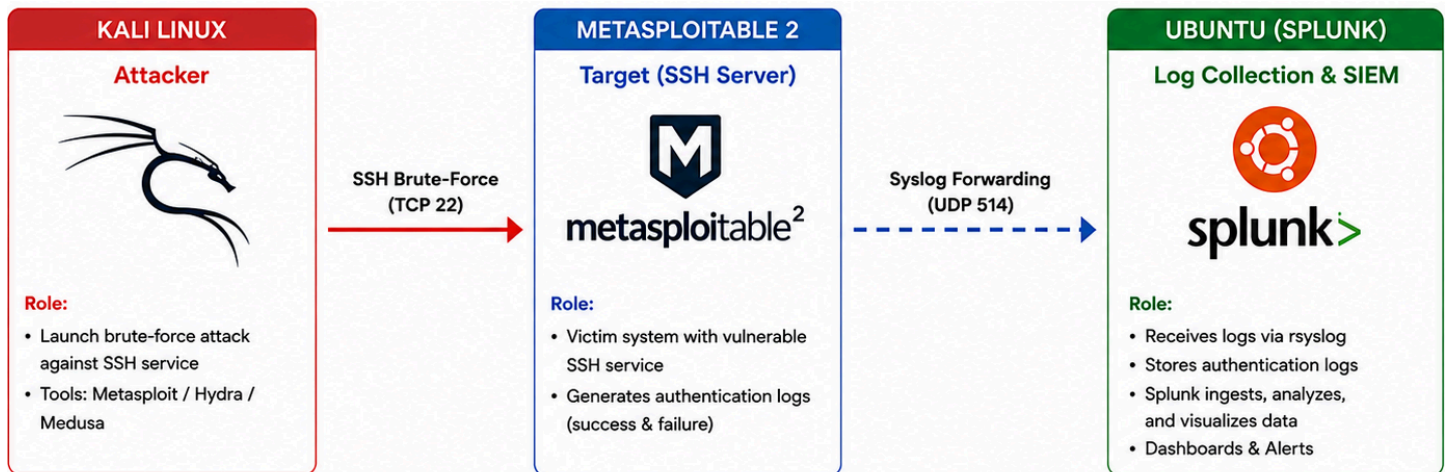
Ubuntu - Log Collection **and** Analysis: Receives the forwarded logs and hosts Splunk for analysis and detection.

Technologies Used

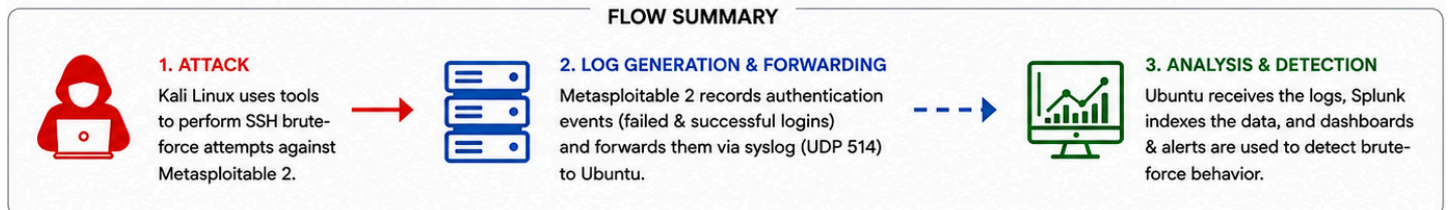
Kali, Metasploitable 2, Ubuntu, VirtualBox, Metasploit, SSH, syslog/rsyslog, Splunk, SPL.

Project Outcomes

Generated controlled SSH password-guessing activity, centralized the resulting logs, built Splunk searches/dashboard visualizations, and created alerts including repeated failures followed by successful authentication.



FLOW SUMMARY



NETWORK
Host-Only Network
(Isolated Lab)



Attack Traffic
(SSH TCP 22)



Log Traffic
(Syslog UDP 514)



GOAL

Detect brute-force SSH attacks by collecting, correlating, and alerting on authentication logs.

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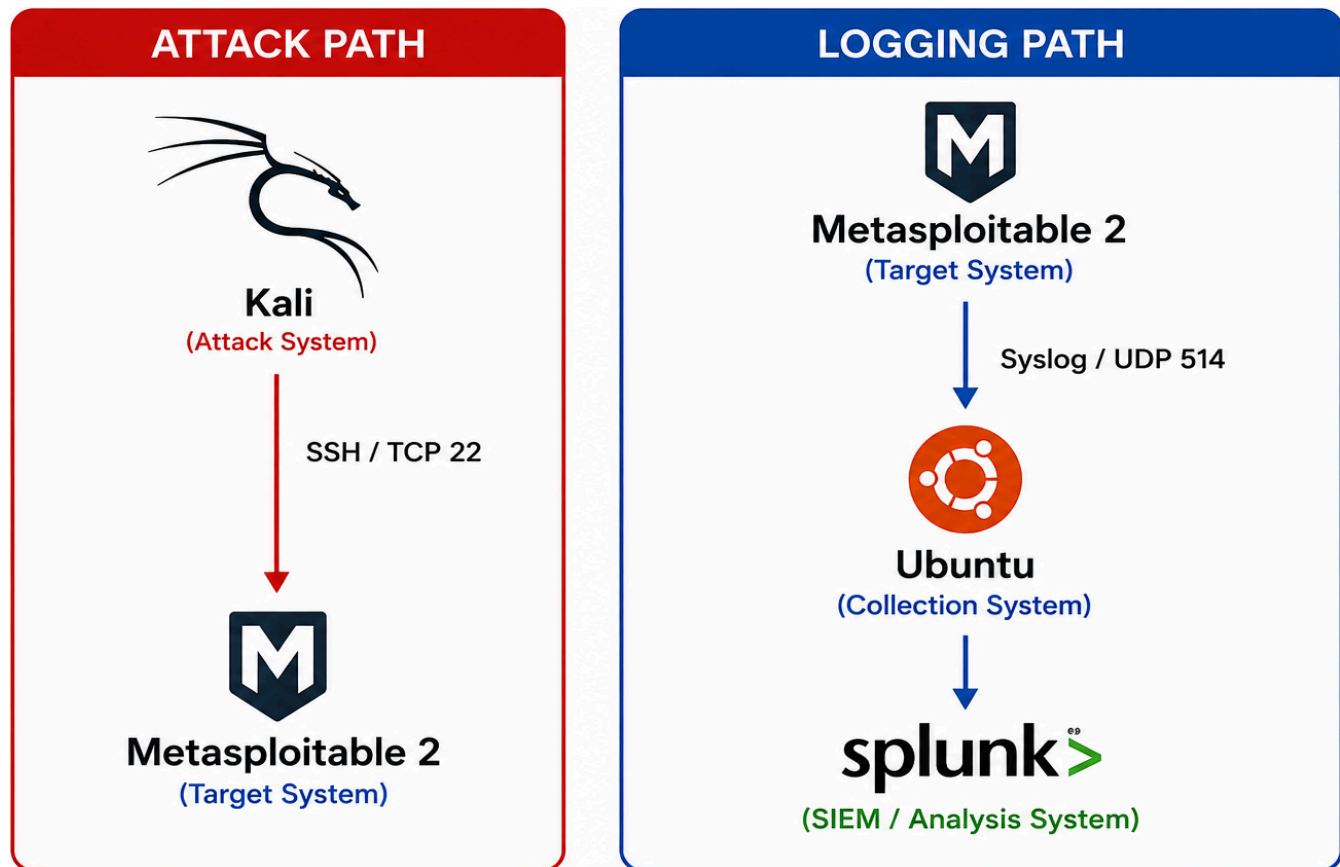
Lab Architecture and Environment

Purpose of the Environment

The lab was designed as an isolated environment where malicious SSH authentication activity could be safely generated, observed, and analyzed. Three virtual machines were assigned separate roles so the attack source, target system, and monitoring infrastructure remained distinct.

VirtualBox Networking

The VMs were configured with a Host-Only network to provide direct communication between the lab systems while keeping the attack environment isolated from external systems. A NAT adapter was also used where Internet connectivity was required for software installation and updates.



Kali - Attack System	Metasploitable 2 - Target	Ubuntu - Log Collection and Analysis
Used to generate controlled SSH password-guessing activity. Metasploit's SSH login scanner was used to automate authentication attempts against the target.	An intentionally vulnerable Linux VM providing the SSH service targeted during the simulation. Authentication attempts generated system logs containing failed and successful login attempts.	Served as the centralized logging system. [rsyslog] received authentication events forwarded by Metasploitable 2, while Splunk ingested the collected logs for searching, visualization, and alerting.

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Log Collection and Analysis

Log Collection Challenge

Metasploitable 2 is an intentionally outdated Linux system and uses a legacy syslog implementation. Rather than relying on a Splunk Universal Forwarder on the target, the lab used the system's existing logging capabilities to remotely forward authentication events to Ubuntu.

Building the Pipeline

1. Authentication Logs Generated

SSH authentication activity against Metasploitable 2 generated local security events, including: Failed and successful authentication attempts, invalid usernames, source IP addresses, and other SSH connection information.

2. Remote Syslog Forwarding

Metasploitable 2's syslog configuration was configured to forward authentication-related events to the Ubuntu collection system using UDP port 514.

```
71:auth,authpriv.* @192.168.56.103:514
```

(Metasploitable2 syslog configuration forwarding authentication events to Ubuntu.)

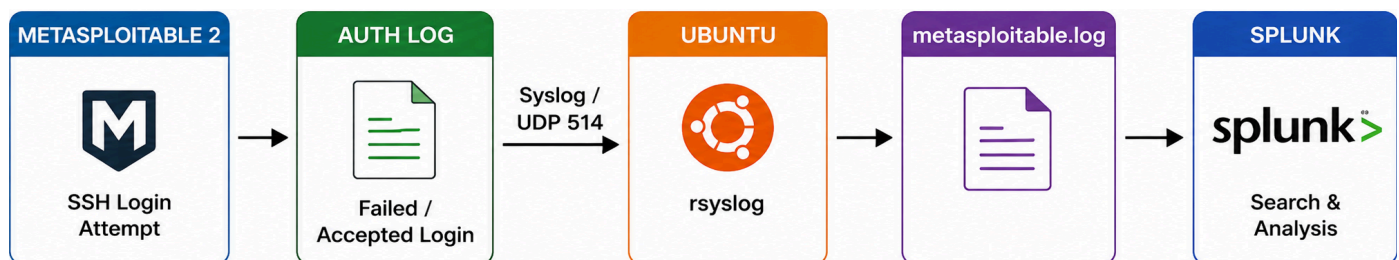
3. Ubuntu Log Collection

Ubuntu's rsyslog service was configured to listen for incoming syslog traffic on UDP 514. Events originating from Metasploitable 2 were written to a dedicated file.

```
osboxes@osboxes:~$ sudo ls -l /var/log/metasploitable.log
-rw-r----- 1 syslog adm 294 Aug 13 12:01 /var/log/metasploitable.log
```

4. Splunk Ingestion

Splunk was configured to monitor the dedicated log file. Once the appropriate file permissions were established, the forwarded authentication events became searchable within Splunk.



Pipeline Validation

Each stage of the pipeline was validated independently. Authentication events were first confirmed locally on Metasploitable 2, network traffic was verified reaching Ubuntu over UDP 514, and the resulting events were confirmed in /var/log/metasploitable.log before configuring Splunk ingestion.

Splunk initially could not access the collected log file because of Linux file permissions. After correcting the permissions and restarting Splunk, the authentication events were successfully indexed and searchable.

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Attack Simulation and Resulting Telemetry

Attack Objective

The attack phase was designed to generate realistic SSH authentication behavior in a controlled environment. Kali Linux was used to perform repeated password-guessing attempts against the SSH service on Metasploitable 2, producing multiple failed authentication events followed by a successful login.

Attack Tool - Metasploit

Metasploit is a penetration-testing framework containing modules for testing systems, services, and vulnerabilities. The auxiliary/scanner/ssh/ssh_login module was used in this lab to automate SSH authentication attempts using a controlled password list.

```
msf auxiliary(scanner/ssh/ssh_login) > run
[*] Starting bruteforce
[*] 192.168.56.102:22 SSH - Testing User/Pass combinations
[-] Failed: 'msfadmin:WrongPassword1'
[!] No active DB -- Credential data will not be saved!
[-] Failed: 'msfadmin:WrongPassword2'
[-] Failed: 'msfadmin:WrongPassword3'
[-] Failed: 'msfadmin:WrongPassword4'
[-] Failed: 'msfadmin:WrongPassword5'
[-] Failed: 'msfadmin:WrongPassword6'
[-] Failed: 'msfadmin:WrongPassword7'
[-] Failed: 'msfadmin:WrongPassword8'
[-] Failed: 'msfadmin:WrongPassword9'
[+] Success: 'msfadmin:msfadmin' 'uid=1000(msfadmin) gid=1000(msfadmin) sshd=1000(msfadmin)'
```

(Metasploit SSH password-guessing output)

Attack Source	Kali Linux
Target	Metasploitable 2
Service	SSH
Port	TCP 22
Tool	Metasploit
Module	auxiliary/scanner/ssh/ssh_login
Attack type	Password guessing
Target account	msfadmin

Resulting Authentication Telemetry

Each authentication attempt caused the SSH service on Metasploitable 2 to generate an authentication event. Failed attempts produced failed-password events, while valid credentials produced an accepted-password event.

```
2026-08-21T13:59:13.343027-04:00 192.168.56.102 sshd[4774]: Failed password for msfadmin from 192.168.56.104 port 33773 ssh2
host = 192.168.56.102 | source = /var/log/metasploitable.log | sourcetype = syslog

2026-08-21T13:59:10.823107-04:00 192.168.56.102 sshd[4771]: Failed password for msfadmin from 192.168.56.104 port 41827 ssh2
host = 192.168.56.102 | source = /var/log/metasploitable.log | sourcetype = syslog

2026-08-21T13:59:08.052681-04:00 192.168.56.102 sshd[4769]: Failed password for msfadmin from 192.168.56.104 port 42975 ssh2
host = 192.168.56.102 | source = /var/log/metasploitable.log | sourcetype = syslog

2026-08-21T13:59:13.904519-04:00 192.168.56.102 sshd[4776]: Accepted password for msfadmin from 192.168.56.104 port 44181 ssh2
host = 192.168.56.102 | source = /var/log/metasploitable.log | sourcetype = syslog
```

(Failed attempts followed by a successful authentication event)

Why this matters

The resulting logs describe the authentication behavior, rather than simply identifying the tool that generated it. Detection logic can therefore focus on patterns such as repeated failures from the same source followed by successful authentication. This allows the detection to identify similar SSH password-guessing behavior regardless of specific tool used to generate attempts.

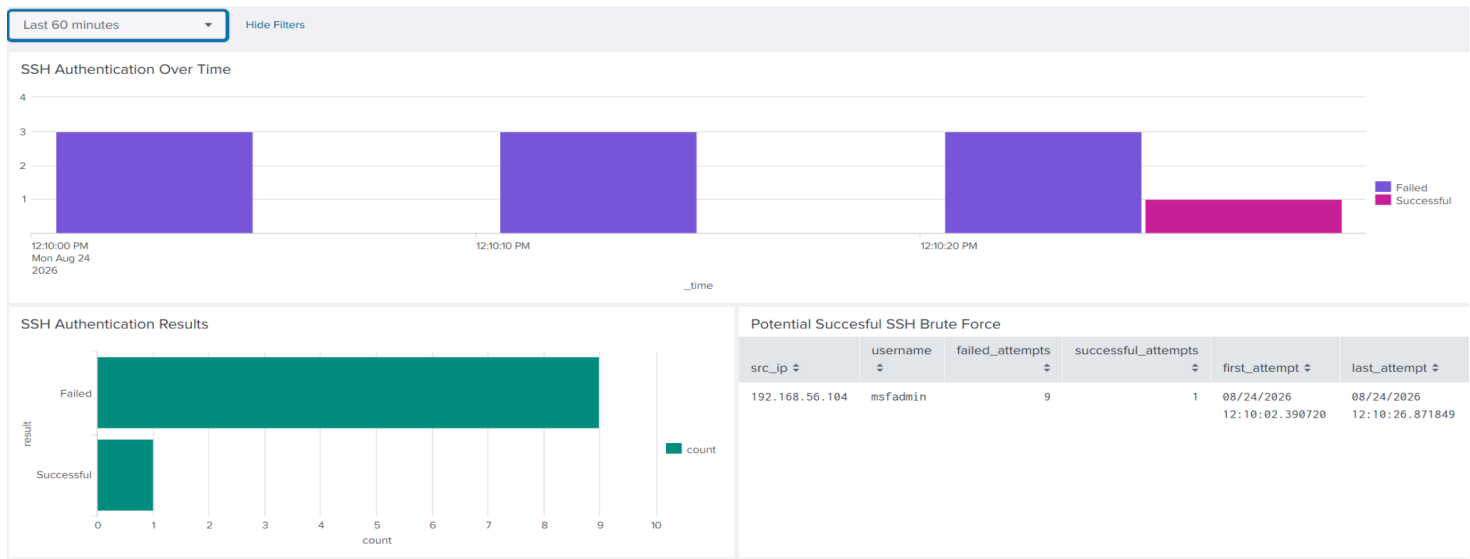
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Splunk Detection and Analysis

Detection Objective

Once the authentication logs were ingested into Splunk, searches were developed to identify and summarize suspicious SSH authentication behavior. Rather than detecting Metasploit specifically, the searches focused on observable behaviors such as repeated failed logins, successful authentication, source IP addresses, and targeted usernames.

Splunk Dashboard



Authentication Over Time	Authentication Results	Potential Brute-Force Success
Displays failed and successful authentication attempts over the selected time period, allowing an analyst to see when the password-guessing activity occurred.	Provides a simple comparison of total failed and successful login attempts.	Highlights suspicious source/user combinations where repeated failures were followed by successful authentication.

Detection Logic

```
index="bruteForce_lab"
("Failed password" OR "Accepted password")
| rex "(?:Failed|Accepted) password for (?<invalid user >)?(?<username>\S+) from (?<src_ip>\d{1,3}(\.?\d{1,3}){3})"
| eval result=case(
  like(_raw,"%Failed password%"), "failure",
  like(_raw,"%Accepted password%"), "success"
)
| stats
  count(eval(result="failure")) AS failed_attempts
  count(eval(result="success")) AS successful_attempts
  max(eval(if(result="failure",_time,null()))) AS last_failure
  max(eval(if(result="success",_time,null()))) AS first_success
by src_ip username
| where failed_attempts >= 5
AND successful_attempts >= 1
AND first_success > last_failure
| convert ctime(last_failure) ctime(first_success)
```

Extract	Source IP and username
Classify	Failure or success
Aggregate	Attempts by source/user
Correlate	Determine whether success followed failures
Filter	Retain activity meeting the suspicious threshold

Detection Result

During the simulated attack, Splunk identified repeated failed authentication attempts against the **msfadmin** account followed by a successful login from the same source. The dashboard summarized the activity and the correlation search identified the sequence as a potential successful SSH brute-force event.

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Automated Detection and Alerting

Alerting Objective

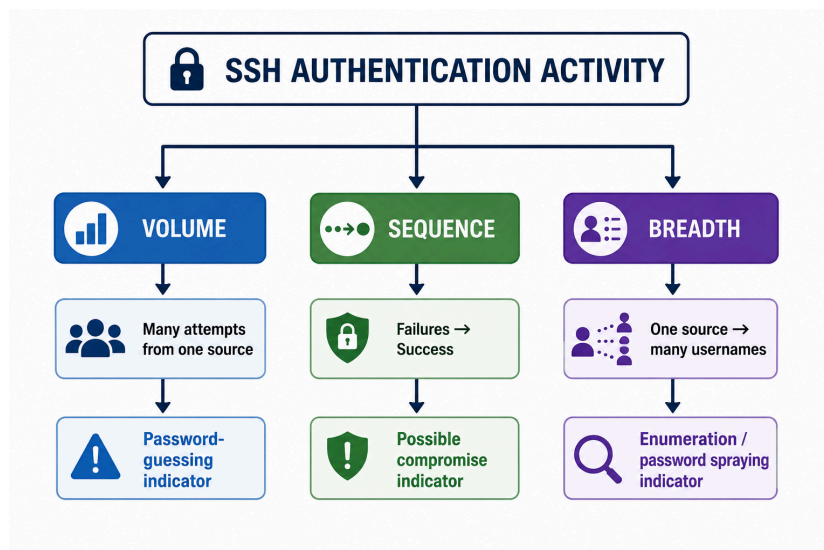
Splunk alerts were created to automatically identify suspicious SSH authentication behavior without requiring an analyst to continuously monitor the dashboard. The alerts use behavioral conditions observed in the authentication logs and escalate from repeated failures to a potentially successful password-guessing attack.

The Three Alerts

Detection	What it Identifies	Why It Matters
Multiple failed SSH login attempts	A source generating multiple SSH login attempts within a defined time period.	Provides an early indication of unusually frequent authentication activity that could represent password guessing.
Multiple Failures Followed by Success	Repeated failed authentication attempts followed by a successful login from the same source against the same account.	Higher-priority behavior because it may indicate that repeated password guessing eventually resulted in valid account access.
One Source IP Attempting Multiple Usernames	A single source IP attempting authentication against multiple usernames.	Can indicate account enumeration, credential spraying, or an attacker searching for a valid account.

3 Distinct Behavioral Perspectives

- Alert 1 - *Volume*. Is one source making an unusual number of authentication attempts?
- Alert 2 - *Sequence*. Did repeated failures eventually lead to a successful authentication?
- Alert 3 - *Breadth*. Is one source attempting to authenticate as multiple different users?



Takeaway

Together, these alerts demonstrate how the same authentication logs can be analyzed from multiple behavioral perspectives. By monitoring volume, sequence, and breadth, Splunk can automatically surface suspicious activity for analyst investigation rather than relying on continuous manual review.

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Conclusions and Lessons Learned

Project Conclusion

This project demonstrated an end to end security monitoring workflow, beginning with controlled SSH password-guessing activity and continuing through log generation, centralized collection, analysis, visualization, and automated detection in Splunk. The completed lab successfully identified several suspicious authentication behaviors, including repeated failures, failures followed by successful authentication, and attempts against multiple usernames by a single source.

Limitations

This was a controlled three-VM lab rather than a production SOC environment. Future development could incorporate additional log sources, stronger SSH security controls, more complex detection scenarios, and automated response mechanisms.

Takeaways

- Detection depends on telemetry. Reliable collection and centralization of authentication logs were necessary before meaningful detection could occur.
- Behavior is more useful than tool identification. The detections focused on authentication patterns rather than identifying Metasploit specifically.
- Correlation provides context. Individual failed logins may be harmless, while repeated failures followed by success represent a much more concerning sequence of events.
- Validation is important. Each stage, from log generation and forwarding to SPL searches and alerts, was tested independently.

Overall Lab Workflow

Simulate attack -> Collect Logs -> Analyze Logs -> Detect Anomalous Behavior -> Alert